



ESTIVA-600
(Efavirenz Tablets 600 mg)

NAME OF THE PRODUCT
ESTIVA-600 (Efavirenz Tablets 600 mg)

NAME AND STRENGTH OF ACTIVE INGREDIENT
Name : Efavirenz
Strength : 600mg

PRODUCT DESCRIPTION
Efavirenz Tablets 600mg

Yellow, capsule shaped, film coated tablets debossed with 'H' on one side and '4' on the other side.

PHARMACODYNAMICS/PHARMACOKINETICS

Pharmacology: Mechanism of Action: Efavirenz is a selective non-nucleoside reverse transcriptase inhibitor of human immunodeficiency virus type 1 (HIV-1). Efavirenz is a non- competitive inhibitor of HIV-1 reverse transcriptase (RT) with respect to template, primer or nucleoside triphosphates, with a small component of competitive inhibition. HIV-2 RT and human cellular DNA polymerases α , β , γ and δ are not inhibited by concentrations of efavirenz.

Pharmacodynamics

Drug Resistance: The potency of efavirenz in cell culture against viral variants with amino acid substitutions at positions 48, 108, 179, 181 or 236 in RT or variants with amino acid substitutions in the protease was similar to that observed against wild type viral strains. The single substitutions which led to the highest resistance to efavirenz in cell culture correspond to a leucine-to-isoleucine change at position 100 (L100I, 17- to 22-fold resistance) and a lysine-to-asparagine at position 103 (K103N, 18- to 33-fold resistance). Greater than 100- fold loss of susceptibility was observed against HIV variants expressing K103N in addition to other amino acid substitutions in RT. K103N was the most frequently observed RT substitution in viral isolates from patients who experienced a significant rebound in viral load during clinical studies of efavirenz in combination with indinavir or zidovudine + lamivudine. Substitutions at RT positions 98, 100, 101, 108, 138, 188, 190 or 225 were also observed, but at lower frequencies, and often only in combination with K103N. The pattern of amino acid substitutions in RT associated with resistance to efavirenz was independent of the other antiviral medications used in combination with efavirenz.

Cross-resistance to Other Antiviral Agents: Cross-resistance profiles for efavirenz, nevirapine and delavirdine in cell culture demonstrated that the K103N substitution confers loss of susceptibility to all 3 NNRTIs. Two of 3 delavirdine-resistant clinical isolates examined were cross-resistant to efavirenz and contained the K103N substitution. A 3rd isolate which carried a substitution at position 236 of RT was not cross-resistant to efavirenz. Viral isolates recovered from PBMCs of patients enrolled in efavirenz who showed evidence of treatment failure (viral load rebound) were assessed for susceptibility to NNRTIs. Thirteen isolates previously characterized as efavirenz-resistant were also resistant to nevirapine and delavirdine. Five of these NNRTI-resistant isolates were found to have K103N or a valine-to- isoleucine substitution at position 108 (V108I) in RT. Three of the efavirenz treatment failure isolates tested remained sensitive to efavirenz in cell culture and were also sensitive to nevirapine and delavirdine. The potential for cross-resistance between efavirenz and protease inhibitors is low because of the different enzyme targets involved. The potential for cross resistance between efavirenz and NRTIs is low because of the different binding sites on the target and mechanism of action.

Cardiac Electrophysiology: The effect of Estiva 600 on the QTc interval was evaluated in an open-label, positive and placebo controlled, fixed single sequence 3-period, 3-treatment crossover QT study in 58 healthy subjects enriched for CYP2B6 polymorphisms. The mean Cmax of efavirenz in subjects with CYP2B6 *6/*6 genotype following the administration of 600 mg daily dose for 14 days was 2.25-fold the mean Cmax observed in subjects with CYP2B6 *1/*1 genotype. A positive relationship between efavirenz concentration and QTc prolongation was observed. Based on the concentration-QTc relationship, the mean QTc prolongation and its upper bound 90% confidence interval are 8.7 ms and 11.3 ms in subjects with CYP2B6*6/*6 genotype following the administration of 600 mg daily dose for 14 days. (See Section Warnings and Precautions & Section Interaction with Other Medicaments)

Pharmacokinetics

Absorption: Dose-related increases in C_{max} and AUC were seen for doses up to 1600 mg; the increases were less than proportional suggesting diminished absorption at higher doses.

Effect of Food on Oral Absorption: The bioavailability of a single 600-mg dose of efavirenz was increased 22% and 17%, respectively, when given with a meal of high fat or normal composition, relative to the bioavailability of a 600-mg dose given under faster conditions. Efavirenz can be administered with or without food.

Distribution: Efavirenz is highly bound (approximately 99.5-99.75%) to human plasma proteins, predominantly albumin.

Metabolism: Efavirenz is principally metabolized by the cytochrome P450 system to hydroxylated metabolites with subsequent glucuronidation of these hydroxylated metabolites. These metabolites are essentially inactive against HIV-1. It is reported that CYP3A4 and CYP2B6 are the major isozymes responsible for efavirenz metabolism. Efavirenz has been shown to induce P450 enzymes, resulting in the induction of its own metabolism.

Elimination: Efavirenz has a relatively long terminal half-life of 52-76 hrs after single doses and 40-55 hrs after multiple doses. Approximately 14-34% of a radiolabeled dose of efavirenz was recovered in the urine and <1% of the dose was excreted in urine as unchanged efavirenz.

INDICATIONS

Efavirenz is indicated in antiviral combination treatment of HIV-1 infected adults, adolescents and children.

RECOMMENDED DOSE

The recommended dosage of Efavirenz in combination with a protease inhibitor and/or nucleoside analogue reverse transcriptase inhibitors (NRTIs) is 600 mg orally, once daily. Efavirenz may be taken with or without food, as desired. In order to improve the tolerability of nervous system side effects, bedtime dosing is recommended during the first 2-4 weeks of therapy and in patients who continue to experience these symptoms.

Concomitant Antiretroviral Therapy: Efavirenz must be given in combination with other antiretroviral medications.

Adolescents and Children (≤ 17 years): The recommended dose of Efavirenz in combination with a protease inhibitor and/or NRTIs for patients ≤ 17 years is described in the table:

Pediatric Dose to be Administered Once Daily	
Body Weight (kg)	Efavirenz Dose (mg)
13 to <15	200
15 to <20	250
20 to <25	300
25 to <32.5	350
32.5 to <40	400
≥ 40	600

Efavirenz tablets should only be administered to children who are able to reliably swallow tablets. Efavirenz tablets may be taken with or without food, as desired. Efavirenz has not been studied in children adequately <3 years or children weighing <13 kg.

Estiva Tablets are available in strength of 600mg. They are not able to deliver all approved dose regimens of Efavirenz; other approved dosage forms and strengths of Efavirenz should be used in such cases

MODE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Efavirenz is contraindicated in patients with clinically significant hypersensitivity to any of its components.

Efavirenz should not be administered concurrently with terfenadine, astemizole, cisapride, midazolam, triazolam or ergot derivatives because competition for CYP3A4 by efavirenz could result in inhibition of metabolism of these drugs and create the potential for serious and/or life-threatening adverse events (eg, cardiac arrhythmias, prolonged sedation or respiratory depression).

Efavirenz must not be administered concurrently with the standard doses of voriconazole because efavirenz significantly decreases voriconazole plasma concentrations while voriconazole also significantly increases efavirenz plasma concentrations.

St. John's wort (*Hypericum perforatum*): Patients on efavirenz should not concomitantly use products containing St. John's wort (*Hypericum perforatum*) since it may be expected to result in reduced plasma concentrations of efavirenz. This effect is due to an induction of CYP3A4 and may result in loss of therapeutic effect and development of resistance.

WARNINGS AND PRECAUTIONS

Efavirenz must not be used as a single agent to treat HIV or added on as a sole agent to a failing regimen.

When prescribing drugs concomitantly with Efavirenz, physicians should refer to the corresponding manufacturer's product circular.

If any antiretroviral medication in a combination regimen is interrupted because of suspected intolerance, serious consideration should be given to simultaneous discontinuation of all antiretroviral medications. The antiretroviral medications should be restarted at the same time upon resolution of the intolerance symptoms. Intermittent monotherapy and sequential reintroduction of antiretroviral agents is not advisable because of the increased potential for selection of drug-resistant mutant virus.

Malformations have been observed in foetuses from efavirenz-treated animals (see Use in pregnancy); therefore, pregnancy should be avoided in women receiving Efavirenz. Barrier contraception should always be used in combination with other methods of contraception (eg, oral or other hormonal contraceptives).

Skin Rash: Mild-to-moderate rash has been reported with Efavirenz and usually resolves with continued therapy. Appropriate antihistamines and/or corticosteroids may improve the tolerability and hasten the resolution of rash. Severe rash associated with blistering, moist desquamation or ulceration has been reported in <1% of patients treated with Efavirenz. The incidence of erythema multiforme or Stevens-Johnson syndrome was also observed. Efavirenz should be discontinued in patients developing severe rash associated with blistering, desquamation, mucosal involvement or fever. If therapy with Efavirenz is discontinued, consideration should also be given to interrupting therapy with other antiretroviral agents to avoid development of drug resistant virus. Rash was reported in children treated with Efavirenz. Prophylaxis with appropriate antihistamines prior to initiating therapy with Efavirenz in children may be considered.

Psychiatric Symptoms: Psychiatric adverse experiences have been reported in patients treated with efavirenz. Patients with a history of psychiatric disorders appear to be at greater risk of these serious psychiatric adverse experiences. There have also been occasional post- marketing reports of death by suicide, delusions and psychosis-like behaviour, although a causal relationship to the use of efavirenz cannot be determined from these reports. Patients should be advised that if they experience these symptoms they should contact their doctor immediately to assess the possibility that the symptoms may be related to the use of efavirenz, and if so, to determine whether the risks of continued therapy outweigh the benefits.

Nervous System Symptoms: Symptoms, including, but not limited to, dizziness, insomnia, somnolence, impaired concentration and abnormal dreaming are frequently reported undesirable effects. Nervous system symptoms usually begin during the first 1 or 2 days of therapy and generally resolve after the first 2-4 weeks. Patients should be informed that if they do occur, these common symptoms are likely to improve with continued therapy and are not predictive of subsequent onset of any of the less frequent psychiatric symptoms. Late-onset neurotoxicity, including ataxia and encephalopathy (impaired consciousness, confusion, psychomotor slowing, psychosis, delirium), may occur months to years after beginning efavirenz therapy. Some events of late-onset neurotoxicity have occurred in patients with CYP2B6 genetic polymorphisms, which are associated with increased efavirenz levels despite standard dosing of Estiva 600. Patients presenting with signs and symptoms of serious neurologic adverse experiences should be evaluated promptly to assess the possibility that these events may be related to efavirenz use, and whether discontinuation of Estiva 600 is warranted.

Seizures: Convulsions have been observed rarely in patients receiving efavirenz, generally in the presence of known medical history of seizures. Patients who are receiving concomitant anticonvulsant medicinal products primarily metabolised by the liver, such as phenytoin, carbamazepine and phenobarbital, may require periodic monitoring of plasma levels. Caution must be taken in any patient with a history of seizures.

Effect of food: The administration of Efavirenz with food may increase Efavirenz exposure and may lead to an increase in the frequency of undesirable effects. This effect may be more evident for the tablets than for the hard capsules. Taking Estiva on an empty stomach, preferably at bedtime, can be considered.

Immune Reconstitution Syndrome: Immune reconstitution syndrome has been reported in patients treated with combination antiretroviral therapy (CART), including Efavirenz. During the initial phase of treatment, a patient whose immune system responds to CART may mount an inflammatory response to indolent or residual opportunistic infections, which may necessitate further evaluation and treatment.

Special Populations: Because of the extensive cytochrome P-450-mediated metabolism of efavirenz and limited clinical experience in patients with chronic liver disease, caution should be exercised in administering Efavirenz to patients with liver disease.

The pharmacokinetics of efavirenz have not been studied in patients with renal insufficiency; however, <1% of an efavirenz dose is excreted unchanged in the urine, so the impact of renal impairment on efavirenz elimination should be minimal. Insufficient numbers of elderly patients have been evaluated in clinical studies to determine whether they respond differently than younger patients.

Efavirenz has not been evaluated in children <3 years or who weigh <13 kg. Evidence exists indicating that Efavirenz may have altered pharmacokinetics in very young children. For this reason, Efavirenz oral solution should not be given to children less than 3 years of age.

Liver Enzymes: In patients with known or suspected history of hepatitis B or C infection and in patients treated with other medications associated with liver toxicity, monitoring of liver enzymes is recommended. In patients with persistent elevations of serum transaminases to >5 times the upper limit of the normal range, the benefit of continued therapy with Efavirenz needs to be weighed against the unknown risks of significant liver toxicity.

Lipids: Monitoring of lipid levels should be considered in patients treated with Estiva.

QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Interaction with Other Medicaments). Consider alternatives to Estiva 600 when co-administered with a drug with a known risk of Torsade de Pointes or when administered to patients at higher risk of Torsade de Pointes.

Effects on ability to drive and use machines

Estiva 600 may cause dizziness, impaired concentration, and/or somnolence. Patients should be

Size : 240 x 350 mm

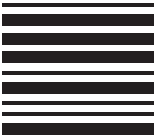
Pharma code: Front- 812 & Back- 813

No.of colors: 01, Black

Note: Position, Height of the pharma code are tentative, it can be changed based on folding size.

Non - Printing Colour :

Die cut



instructed that if they experience these symptoms they should avoid potentially hazardous tasks such as driving or operating machinery.

DRUG INTERACTIONS

Efavirenz is an inducer of CYP3A4. Other compounds that are substrates of CYP3A4 may have decreased plasma concentrations when co-administered with Efavirenz.

Amprenavir: While clinical significance of decreased amprenavir concentrations has not been established, it should be taken into consideration when choosing a regimen containing both efavirenz and amprenavir.

Fosamprenavir calcium: For coadministration with fosamprenavir and ritonavir, the prescribing information for fosamprenavir calcium should be consulted.

Atazanavir: Coadministration of efavirenz 600mg with atazanavir resulted in substantial decreases in atazanavir exposure, necessitating dosage adjustment of atazanavir in combination with ritonavir. Efavirenz concentrations when co-administered with atazanavir and low-dose ritonavir were similar to efavirenz alone, based on comparision with historical data.

Indinavir: The optimal dose of indinavir, when given in combination with Efavirenz, is not known. Increasing the indinavir dose to 1000mg every 8 hours does not compensate for the increased indinavir metabolism due to efavirenz.

Ritonavir: When Efavirenz and ritonavir was administered in uninfected volunteers, the combination was not well tolerated and was associated with a higher frequency of adverse clinical experiences (eg, dizziness, nausea, paresthesia) and laboratory abnormalities (elevated liver enzymes). Monitoring of liver enzymes is recommended when Efavirenz is used in combination with ritonavir.

Saquinavir: Use of Efavirenz in combination with saquinavir as the sole protease inhibitor is not recommended.

Rifamycins: Rifampin reduced efavirenz concentration in uninfected volunteers. The dose of Efavirenz should be increased to 800mg/day when taken with rifampin. No dose adjustment of rifampin is recommended when given with Efavirenz. Rifabutin had no significant effect on the pharmacokinetics of efavirenz. These data suggest that the daily dose of rifabutin should be increased by 50% when administered with efavirenz and that the rifabutin dose may be doubled for regimens in which rifabutin is given 2 or 3 times a week in combination with efavirenz.

Clarithromycin: No dose adjustment of Efavirenz is recommended when given with clarithromycin. Alternatives to clarithromycin should be considered.

Voriconazole: Co-administration of Efavirenz and voriconazole is contraindicated.

Itraconazole: Since no dose recommendation for itraconazole can be made, alternative antifungal treatment should be considered.

Lipid-lowering agents: Coadministration of efavirenz with the HMG-CoA reductase inhibitors atorvastatin, pravastatin, or simvastatin has been shown to reduce the plasma concentration of the statin in uninfected volunteers. Cholesterol levels should be periodically monitored. Dosage adjustments of statins may be required. Coadministration of efavirenz with atorvastatin, pravastatin, or simvastatin did not affect efavirenz AUC or Cmax values. No dosage adjustment is necessary for efavirenz.

Carbamazepine: Carbamazepine plasma levels should be monitored periodically. There are no data with coadministration or higher doses of either medicinal product; therefore, no dose recommendation can be made, and alternative anticonvulsant treatment should be considered.

Other anticonvulsants: No data are available on the potential interactions of efavirenz with phenytoin, phenobarbital, or other anticonvulsants that are substrates of CYP450 isozymes. When efavirenz is administered concomitantly with these agents, there is a potential for reduction or increase in the plasma concentrations of each agent; therefore, periodic monitoring of plasma levels should be conducted. Clinically significant interactions would not be expected since vigabatrin and gabapentin are exclusively eliminated unchanged in the urine and would be unlikely to compete for the same metabolic enzymes and elimination pathways as efavirenz.

Oral Contraceptives: Because the potential interaction of efavirenz with oral contraceptives has not been fully characterized, a reliable method of barrier contraception should be used in addition to oral contraceptives.

Methadone: In HIV-infected IV drug users, co-administration of efavirenz with methadone resulted in decreased plasma levels of methadone and signs of opiate withdrawal. The methadone dose was increased to alleviate withdrawal symptoms. Patients should be monitored for signs of withdrawal and their methadone dose increased as required to alleviate withdrawal symptoms.

St. John's Wort (Hypericum perforatum): Patients on efavirenz should not concomitantly use products containing St. John's wort (Hypericum perforatum) since it may be expected to result in reduced plasma concentrations of efavirenz. This effect is due to an induction of CYP3A4 and may result in loss of therapeutic effect and development of resistance.

Antidepressants: No dose adjustments are necessary for either efavirenz of paroxetine when these drugs are co-administered. Sertraline did not significantly alter the pharmacokinetics of efavirenz. The dose of sertraline should be increased when administered with efavirenz to compensate for the induction of sertraline metabolism by efavirenz. Sertraline dose increases should be guided by clinical response.

Cetirizine: No dose adjustments are necessary for either efavirenz or cetirizine when these drugs are co-administered.

Lorazepam: No dose adjustments are necessary for either efavirenz or lorazepam when these drugs are co-administered.

Calcium channel blockers: No data are available on the potential interactions of efavirenz with other calcium channel blockers that are substrates of the CYP3A4 enzyme (eg, verapamil, felodipine, nifedipine, nicardipine). When efavirenz is administered concomitantly with one of these agents, there is a potential for reduction in the plasma concentrations ot the calcium channel blocker. Dose adjustments should be guided by clinical response (refer to the corresponding manufacturer's product circular for the calcium channel blocker).

Cannabinoid Test Interaction: Efavirenz does not bind to cannabinoid receptors. False- positive urine cannabinoid test results have been reported.

QT Prolonging Drugs: There is limited information available on the potential for a pharmacodynamic interaction between Estiva 600 and drugs that prolong the QTc interval. QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Warnings and Precautions). Consider alternatives to Estiva 600 when co-administered with a drug with a known risk of Torsade de Pointes.

PREGNANCY AND LACTATION PREGNANCY

There are no adequate and well-controlled studies of efavirenz in pregnant women. Pregnancy should be avoided in women treated with efavirenz. Barrier contraception should always be used in combination with other methods of contraception (for example, oral or other hormonal contraceptives). Because of the long half life of efavirenz, use of adequate contraceptive measures for 12 weeks after discontinuation of Efavirenz is recommended. Women of childbearing potential should undergo pregnancy testing before initiation of efavirenz. Efavirenz should not be used during pregnancy unless the potential benefit to the mother clearly outweighs the potential risk to the fetus and there are no other appropriate treatment

options. If a woman takes efavirenz during the first trimester of pregnancy or becomes pregnant while taking efavirenz, she should be informed of the potential harm to the fetus.

NURSING MOTHERS

Efavirenz is secreted into the milk of lactating rats and efavirenz also has been shown to pass into human breast milk. It is recommended that mothers taking efavirenz do not breast-feed their infants. It is recommended that HIV-infected women do not breast-feed their infants under any circumstances in order to avoid transmission of HIV.

UNDESIRABLE EFFECTS

Efavirenz was generally well tolerated. In patients who received efavirenz in combination with protease inhibitors and/or NRTIs, the most frequently reported treatment-related undesirable effects were rash, dizziness, nausea, headache and fatigue. The most notable undesirable effects associated with efavirenz are rash and nervous system symptoms.

Other undesirable effects include: Allergic reaction, abnormal coordination, ataxia, confusion, stupor, vertigo, vomiting, diarrhoea, hepatitis, impaired concentration, insomnia, anxiety, abnormal dreams, somnolence, depression, abnormal thinking, agitation, amnesia, delirium, emotional lability, euphoria, hallucination and psychosis.

Additional undesirable effects reported in post-marketing surveillance include neurosis, paranoid reaction, cerebellar coordination and balance disturbances, convulsions, pruritus, abdominal pain, blurred vision, flushing, gynecomastia, hepatic failure, photoallergic dermatitis, pancreatitis, redistribution/accumulation of body fat in areas such as the back of the neck, breasts, abdomen, and retroperitoneum, tinnitus, and tremor.

A few of the post-marketing reports of hepatic failure, including cases in patients with no pre-existing hepatic disease or other identifiable risk factors, were characterized by a fulminant course, progressing in some cases to transplantation or death.

The type and frequency of undesirable effects in children were generally similar to that of adult patients with the exception that rash was reported more frequently in children and was more often of higher grade than in adults.

Rash: Skin rash was considered treatment-related in patients treated with Efavirenz. The incidence of erythema multiforme or Stevens-Johnson syndrome was also reported. Prophylaxis with appropriate antihistamines prior to initiating therapy with efavirenz in children may be considered. Rashes are usually mild-to-moderate maculopapular skin eruptions that occur within the first 2 weeks of initiating therapy with Efavirenz. In most patients, rash resolves with continuing therapy with Efavirenz within 1 month. Efavirenz can be re-initiated in patients interrupting therapy because of rash. Use of appropriate antihistamines and/or corticosteroids is recommended when Efavirenz is restarted.

Psychiatric Symptoms: Serious psychiatric adverse experiences have been reported in patients treated with efavirenz. Symptoms include severe depression, suicidal ideation, nonfatal suicide attempts, aggressive behaviour, paranoid reactions and manic reactions. Patients with a history of psychiatric disorders appear to be at greater risk of these serious psychiatric adverse experiences, with the frequency of each of the previously mentioned events ranging from manic reactions to both severe depression and suicidal ideation. There have also been occasional reports of death by suicide, delusions and psychosis-like behavior, although a causal relationship to the use of efavirenz cannot be determined from these reports.

Nervous System Symptoms: Symptoms including, but not limited to, dizziness, insomnia, somnolence, impaired concentration, and abnormal dreaming are frequently reported side effects in patients receiving Efavirenz 600 mg daily. Nervous system symptoms usually begin during the first 1 or 2 days of therapy and generally resolve after the first 2-4 weeks. Dosing at bedtime improves the tolerability of these symptoms and is recommended during the first weeks of therapy and in patients who continue to experience these symptoms. Dose reduction or splitting the daily dose has not been shown to provide benefit and is not recommended.

Postmarketing experiences: Encephalopathy.

SYMPTOMS AND TREATMENT OF OVERDOSE

Some patients accidentally taking 600 mg twice daily have reported increased nervous system symptoms such as involuntary muscle contractions. Treatment of overdose with Efavirenz should consist of general supportive measures, including monitoring of vital signs and observation of the patient's clinical status. Administration of activated charcoal may be used to aid removal of unabsorbed drug. There is no specific antidote for overdose with Efavirenz. Since efavirenz is highly protein-bound, dialysis is unlikely to significantly remove the drug from blood.

STORAGE CONDITION

Store below 30°C and protect from moisture.

DOSAGE FORM AND PACKAGING AVAILABLE.

30 Tablets in HDPE container.

PRODUCT REGISTRATION HOLDER:

CAMBER LABORATORIES SDN BHD
Unit E-13A-02, Menara SUEZCAP 2, No. 2, KL Gateway,
Jalan Kerinchi, Gerbang Kerinchi Lestari,
59200 Kuala Lumpur, Malaysia

MANUFACTURER

HETERO LABS LIMITED

Unit V, Sy. No 439,440,441 & 458
TSIIC Formulation SEZ, Polepally Village,
Jadcherla Mandal, Mahaboobnagar Dist.
Telangana, India

DATE OF REVISION

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